Decision Models for Medical Decision Support

1 Context of the internship

The internship will take place at Télécom Paris, within the Image, Données, Signal (IDS) Department, in the IMAGES research team. It will start in spring 2027 for four to six months, in close interaction with researchers and PhD students working on machine learning for biomedical data and medical imaging.

2 Internship issue

Many clinical decisions are choices among a few known options: is a finding present? Which grade is it? Should the patient be referred? For such decisions, clinicians need not only an answer but a confidence they can trust: a probability that matches how often the model is right, so that confident cases can be handled automatically and uncertain ones referred to a clinician. Large language models (LLMs) answer by generating free text: the answer may fall outside the valid options or contain invented content, and its confidence reflects the likelihood of words rather than of the decision itself [1].

Decision models are a new family of AI models designed for this setting. Given an input and a question with a fixed list of possible answers, they return a probability for each option and generate no text: they cannot invent an answer outside the options, and they are trained so that their probabilities are calibrated, i.e. an answer given with 80% confidence should be right about 80% of the time. The first one, Jev, was released in September 2026 [2]; open reproductions have followed [3], some of which accept multi-modalities [4], and a medical decision model, OmniMed-Jev, reports much more calibrated probabilities than its text-generating counterpart [1].

Decision models are not the only methods that output such probabilities. For clinical data combining medical images and tabular patient variables, tabular foundation models (TFMs) such as TabPFN [5] and TabICL [6], combined with image features, form strong vision–tabular methods. They predict from labelled patients given as context, whereas decision models rely on the meaning of the question and options and can answer with few or no labels.

The central question of this internship is therefore: **on clinical image and tabular data, which approach gives the most trustworthy confidence: decision models, or vision–tabular methods based on TFMs?** Here, a confidence is trustworthy if it matches how often the model is right, if it is lower on wrong decisions, and if both properties hold when the data change.

3 Analysis of the current situation

The confidence of decision models is claimed, not verified on clinical data. Open decision models have never been evaluated on clinical image–tabular data. OmniMed-Jev was evaluated on images only, without patient variables, and on data from the same sources as its training data [1]. Confidence may also break when the data change: for Jev, a threshold tuned on one dataset wrongly rejected 69.3% of valid inputs on another [7].

No comparison with tabular foundation models. No study compares the confidence of decision models and TFMs on the same clinical tasks and with the same labelled patients.

4 Scientific proposal

Step 1: comparing the confidence of the two families. On public clinical image–tabular datasets, the intern will compare (i) open decision models, both general multimodal ones and OmniMed-Jev, and (ii) vision–tabular methods feeding image features and clinical variables to TabPFN or TabICL. For a fair comparison, both families answer the same classification tasks, with the same classes, on the same test patients, and use the same labelled patients: as context for TFMs, and to adjust the decision models, which are also evaluated without labels. Since both families output a probability for each class, their confidence (the probability of the chosen answer) can be compared directly. Besides accuracy, we will measure:

- *calibration*: does a confidence of 80% mean that the decision is right 80% of the time? (ECE, Brier score, reliability diagrams [8]);
- *error detection*: is the confidence lower on wrong decisions? (error rate when only the most confident cases are answered and the others referred to a clinician);
- *robustness*: do these properties hold on data from another source or with missing variables?

Step 2: adapting decision models to the medical setting. The intern will then study how to make the confidence of decision models more trustworthy in clinical use, using the same labelled patients as the TFMs, and compare the adapted models again with TFMs. Possible directions include: designing the question, options and tabular inputs, without training; recalibrating the probabilities with a few labelled patients [8]; and lightly fine-tuning a small decision model with LoRA [9], as in OmniMed-Jev [1].

5 Expected results

- A reliability benchmark of decision models and TFM-based methods on clinical data, with reusable code.
- A report on which approach gives the most trustworthy decisions, and under which conditions.
- An adapted decision model for medical data. Depending on the results, a workshop or conference paper.

How to apply?

- Contact:`shurong.zhang@telecom-paris.fr` and `maxime.difolco@telecom-paris.fr`
- Documents to include in the application: resume, transcripts.

References

[1] Luyao Tang and Cheng Chen. *OmniMed-Jev: Calibrating LVLM Confidence for Trustworthy Medical Multimodal Decisions via System One*. 2026. arXiv: `2610.00381 [cs.LG]`. URL: `https://arxiv.org/abs/2610.00381`.

[2] TypeSafe AI. *Jev: A Decision Model*. Released in September 2026. Documentation on OpenRouter. 2026. URL: `https://openrouter.ai/docs/guides/community/jev`.

[3] Xialie Zhuang et al. *StartLux-Decision: AutoThink Decision Models with Reinforcement Learning for Calibrated Deliberative Decisions*. 2026. arXiv: `2610.17309 [cs.CL]`. URL: `https://arxiv.org/abs/2610.17309`.

[4] OmniJev Team. *OneJev-27B: A Multimodal System One Decision Model*. Hugging Face model card and GitHub repository. 2026. URL: `https://huggingface.co/OmniJev/OneJev-27B`.

[5] Noah Hollmann et al. "Accurate predictions on small data with a tabular foundation model". In: *Nature* 637 (2025), pp. 319–326. URL: `https://doi.org/10.1038/s41586-024-08328-6`.

[6] Jingang Qu et al. *TabICL: A Tabular Foundation Model for In-Context Learning on Large Data*. 2025. arXiv: `2502.05564 [cs.LG]`. URL: `https://arxiv.org/abs/2502.05564`.

[7] Jiawen Lu and Tongtong Wu. *Benchmarking Candidate Coverage and Rejection Policy Transfer in Typed Decision Models*. 2026. arXiv: `2610.03387 [cs.AI]`. URL: `https://arxiv.org/abs/2610.03387`.

[8] Chuan Guo et al. "On Calibration of Modern Neural Networks". In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. 70. Proceedings of Machine Learning Research. PMLR, 2017, pp. 1321–1330. URL: `https://proceedings.mlr.press/v70/guo17a.html`.

[9] Edward J. Hu et al. "LoRA: Low-Rank Adaptation of Large Language Models". In: 2021. arXiv: `2106.09685 [cs.CL]`. URL: `https://arxiv.org/abs/2106.09685`.